Lakshay Chauhan

(+91) 99114 30026 | lakshay@9th.fun | https://9th.fun | linkedin.com/in/nos1dot618 | github.com/nos1dot618

Education

Indraprastha Institute of Information Technology (IIIT) Delhi Jul 2021 – Jun 2025
Bachelor of Technology in Computer Science and Engineering *Delhi, India*

Work Experience

Member of Technical Staff Jul 2025 – Present
ZL Technologies, Inc | ***Website*** *Hyderabad, India*

- Developed task synchronization and bulk parallel processing mechanisms across clusters within ZL's analytics module, achieving over 100% performance improvement.
- Implemented task completion time estimation and duplicate item detection for large-scale data loading workflows processing millions of items.

Research Assistant | ***Gitlab Project Organization*** May 2024 – May 2025
Networks and Systems Security Lab - IIITD | ***Website*** *Delhi, India*

- Researched mechanisms to mitigate video streaming quality degradation during fallback from QUIC to TCP caused by UDP blocking.
- Implemented QUIC-based notification signaling and kernel congestion window state reuse to bypass TCP slow start, reducing buffering and improving user experience during fallback transitions.

Projects

Task-Based Parallel Runtime Library | ***Repository*** Jan 2025 – Apr 2025

- **Key Skills:** Parallel Programming, Runtime Systems, C++, Multithreading, NUMA
- Designed a task-based parallel runtime using a worker-based model to minimize synchronization overhead.
- Developed async-finish, energy-efficient, NUMA-aware, receiver-initiated, and trace-and-replay runtimes to study scalability and memory locality.

Minimal Compiler Infrastructure | ***Repository*** May 2024 – Present

- **Key Skills:** Compiler Infrastructure, Optimization, Rust, FASM, WASM, Graphviz
- Developed a minimal LLVM-inspired compiler that converts source code into optimized CFGs and target-specific assembly.
- Implemented CFG optimization passes including identifier validation, constant folding, and CFG simplification.

TCP Congestion Window Analysis with eBPF | ***Repository*** Nov 2024 – Dec 2024

- **Key Skills:** C/C++, Linux Kernel, eBPF, TCP/IP, Systems Programming
- Developed eBPF programs to observe and modify TCP congestion windows via kernel hooks and user-space BPF maps.
- Implemented kernel-user space interaction using libbpf and bpftool for runtime control, tracing, and performance analysis.

Open Source

Active open-source contributor to Microsoft, PowerShell, Nushell, and other projects, with contributions spanning features, bug fixes, developer tooling, documentation, and code reviews.

Technical Skills

Languages: C, C++, Rust, Assembly, Java, Kotlin, JavaScript, Python, Bash, Haskell
Frameworks: PyTorch, Django, ReactJS, Numpy, Pandas, Tauri, LibGDX, Raylib
Developer Tools: Emacs, Linux, Git, GDB, Markdown, Google Cloud Platform, OpenLiteSpeed, SqlLite3

Awards

- Awarded the **Summer Undergraduate Research Fellowship** (2023) by **IRD-IIITD** for the project *"Utilizing Ultrasonic Distance Sensors as a Mapping Tool to Design User-Friendly CST"*.
- Awarded the **CHANAKYA Fellowship** (2024) by **iHub Anubhuti Foundation** for the project *"A Unified Approach to User Emotion Detection through Emojis and Textual Analysis"*.